Internet Appendix for

Firm-Level Labor-Shortage Exposure

Jarrad Harford, Qiyang He, and Buhui Qiu

Appendix A

Table A1. Variable Definitions

Variables	Definition
Dependent Variables	
Unemployment Rate	A state's total number of people unemployed divided by the state's total labor force in a year. **Source:** U.S. Bureau of Labor Statistics (BLS).
Unemployment Rate^{ind}	An industry's (3-digit NAICS) total number of people unemployed divided by the total labor force in a year. **Source:** Current Population Survey (CPS) from U.S. Bureau of Labor Statistics (BLS).
Wage Growth	A state's total wage in year t minus the state's total wage in year t-1, further divided by the state's total wage in year t-1. **Source:** U.S. Bureau of Labor Statistics (BLS).
Wage Growth^{ind}	An industry's (3-digit NAICS) total wage in year t minus the total wage in year t-1, further divided by the industry's total wage in year t-1. **Source:** Quarterly Census of Employment and Wages (QCEW) from U.S. Bureau of Labor Statistics (BLS).
Log($\frac{\#Job\ Openings}{\#Unemployed}$)	Natural logarithm of a state's number of job openings divided by the state's number of people unemployed in a year. **Source:** U.S. Bureau of Labor Statistics (BLS).
Log($\frac{\#Job\ Openings}{\#Unemployed}$)^{ind}	Natural logarithm of an industry's (3-digit NAICS) number of job openings divided by the number of people unemployed in a year. JOLTS provides job openings data only at the 2-digit NAICS level. As such, to estimate the job openings data at 3-digit NAICS level, we scale the 2-digit NAICS job openings data by the employment share that each 3-digit industry accounts for within its corresponding 2-digit industry. **Source:** Job Openings and Labor Turnover Survey (JOLTS) from U.S. Bureau of Labor Statistics (BLS).
CAR(0, 2)	Cumulative abnormal stock returns within a three-day event window of (0, 2) following the earnings conference calls. **Source:** CRSP
Growth in Per-Employee Staff Expenses	The difference in the natural logarithm of a firm's per-employee staff expenses (XLR/EMP) between year *t* and year *t-1*. If a firm does not report total staff expenses in a year, we complement it with estimated total staff expenses, which equals to the firm's selling, general & administrative expenses (XSGA) in the year multiplied by the industry-year average ratio of $\frac{XLR}{XSGA}$. **Source:** Compustat.
Stock Return	Buy-and-hold stock return of a firm. **Source:** CRSP
FF3F-adjusted Stock Return	The Fama-French three-factor-adjusted stock returns. **Source:** CRSP and Kenneth R. French Data Library
FF5F-adjusted Stock Return	The Fama-French five-factor-adjusted stock returns. **Source:** CRSP and Kenneth R. French Data Library

ROA	A firm's earnings before extraordinary items divided by the book value of assets (OIBDP/AT). **Source:** Compustat.
Ind-adj. ROA	A firm's operating income before depreciation divided by the firm's book value of total assets, further adjusted by the average ROA of firms in the same industry (2-digit SIC) and year. **Source:** Compustat.
OCF	a firm's operating cash flow divided by the firm's book value of total assets. (OANCF/AT). **Source:** Compustat.
Ind-adj. OCF	A firm's operating cash flow divided by the firm's book value of total assets, further adjusted by the average OCF of firms in the same industry (2-digit SIC) and year. **Source:** Compustat.
CAPEX	A firm's capital expenditures divided by the book value of assets (CAPX/AT). **Source:** Compustat.
R&D	A firm's research and development expenses divided by the book value of total assets (XRD/AT). Missing XRD is replaced with zero. **Source:** Compustat.
△Employees/AT	The change in a firm's number of employees (in thousand) divided by the book value of assets (EMP/AT). **Source:** Compustat.
△AI Employee Share	The change in a firm's AI employee share from year *t-1* to year *t*. **Source: ?**.
Process Claims Share	The number of process claims divided by the number of total claims for all patents a firm has applied for (and later granted) in year t. **Source: ?** process patent classification dataset.
#CW Process Patents/AT	The citation weighted number of process patents a firm has applied (and later granted), further scaled by the total assets of the firm at the beginning of the year. **Source: ?** process patent classification dataset.
#CW Non-Process Patents/AT	The citation weighted number of non-process patents a firm has applied (and later granted), further scaled by the total assets of the firm at the beginning of the year. **Source: ?** process patent classification dataset.
#CW AI Patents/AT	The citation weighted number of AI patents a firm has applied (and later granted), further scaled by the total assets of the firm at the beginning of the year. **Source:** USPTO official website
Log(Per Employee Staff Expenses)	The natural logarithm of a firm's per-employee staff expense (XLR/EMP) in a year. If a firm does not report total staff expenses in a year, we complement it with estimated total staff expenses, which equals to the firm's selling, general & administrative expenses (XSGA) in the year multiplied by the industry-year average ratio of $\frac{XLR}{XSGA}$. **Source:** Compustat.
Employees/AT	A firm's number of employees (in thousand) divided by the book value of assets in a year (EMP/AT). **Source:** Compustat.
Independent Variables	
LS Exposure^{state}	The average labor-shortage exposure of all public firms headquartered in that state in a year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
LS Exposure^{ind}	The labor-shortage exposure at 3-digit NAICS industry level, measured as the average labor-shortage exposure across firms by 3-digit NAICS level in a year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
LS Exposure	The average number of labor-shortage-related sentences divided by the average number of total sentences of earnings call transcripts of a firm in a year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.

I (LS)	An indicator variable that equals one if the *LS Exposure* of a firm in that year is larger than zero, and equals zero otherwise. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
#LS Sentences	The number of labor-shortage sentences of earnings call transcripts of a firm in a year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
Log(1+#LS Sentences)	The natural logarithm of one plus the number of labor-shortage sentences of earnings call transcripts of a firm in a year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
$\frac{\#LS\ Sentences}{\#Labor\ Sentences}$	The average number of labor-shortage sentences divided by the average number of labor-related sentences of earnings call transcripts of a firm in a year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
LS Exposure^{SW}	Severity-weighted labor shortage exposure, measured as the severity-weighted sentence count scaled by total sentence count and averaged across all conference calls in the year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
LS Exposure^{Positive}	The average number of positive labor-shortage sentences divided by the average number of total sentences of earnings call transcripts of a firm in a year. The sentiment is predicted using *FinBert* model trained by **?**. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
LS Exposure^{NonPositive}	The average number of non-positive labor-shortage sentences divided by the average number of total sentences of earnings call transcripts of a firm in a year. The sentiment is predicted using *FinBert* model trained by **?**. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
LS Exposure^{Mgmt}	The average number of labor-shortage sentences divided by the average number of total sentences in the presentation section of earnings call transcripts of a firm in a year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
LS Exposure$^{\text{Q\&A}}$	The average number of labor-shortage sentences divided by the average number of total sentences in the Q&A section of earnings call transcripts of a firm in a year. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
MA3-$\triangle$LS^{Increase}	The change of 3-year moving average of *LS Exposure* between year *t* and *t-1* when it is positive, and zero otherwise. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
MA3-$\triangle$LS^{Decrease}	The change of 3-year moving average of *LS Exposure* between year *t* and *t-1* when it is negative, and zero otherwise. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
Log(GDP)	Natural logarithm of annual GDP of a state in a year. **Source:** Bureau of Economic Analysis (BEA)
Log(Population)	Natural logarithm of total population of a state in a year. **Source:** Bureau of Economic Analysis (BEA)
Log(Per Cap Income)	Natural logarithm of per capital income of a state in a year. **Source:** Bureau of Economic Analysis (BEA)
Industry Size	Natural logarithm of the average sales (SALE) of 3-digit NAICS industry firms in a year. **Source:** Compustat.
Firm Size	Natural logarithm of the sales (SALE) of a firm in a year. **Source:** Compustat.
Earnings Surprise	Actual quarterly earnings per share (EPS) announced in a quarter minus median analyst forecasted EPS made before the EPS announcement quarter, scaled by absolute stock price at the end of the quarter before the EPS announcement quarter. **Source:** I/B/E/S and CRSP
R&D Stock	The accumulated R&D expenses (XRD) of the firm over the five-year period ending in year t-2, assuming an annual depreciation rate of 20% following **?**. **Source:** Compustat.

△CAPEX	The change in a firm's capital expenditure from year *t-1* to year t, divided by the firm's capital expenditure in year *t-1*. **Source:** Compustat.
△R&D	The change in a firm's R&D expenses from year *t-1* to year t, divided by the firm's R&D expenses in year *t-1*. **Source:** Compustat.
△CW Process Patent	The change in a firm's number of citation-weighted process patents from year *t-1* to year *t*, divided by the firm's number of citation-weighted process patents in year *t-1*. **Source: ?** process patent classification dataset.
Monopsony Power (1/0)	An indicator variable that equals one if a firm operates in a state-industry (SIC2)-year with an above-median employment-based Herfindahl-Hirschman Index (HHI) and itself ranks above the median in employment size, otherwise equals zero. **Source:** Compustat.
Repeated LS (1/0)	An indicator variable that equals one if a firm has experienced labor shortages at least once during the three years prior to the current LS exposure, otherwise equals zero. **Source:** S&P Capital IQ and the fine-tuned machine learning model.
Financial Constraints (1/0)	An indicator variable that equals one if at least three out of four financial constraint measures (i.e., the **?** index, the **?** debt constraint index, the **?** equity constraint index, and cash dividend payment indicator) suggest the firm as financially constraints, otherwise equals zero. **Source: ?**, **?**, Compustat

Table A3. Summary Statistics for Variables Used in Appendix A

This table reports the summary statistics for the variables used in Appendix A. We report the number of observations, mean, 25th percentile, median, 75th percentile, and standard deviation. All financial variables are winsorized at the 1st and 99th percentiles. Table A1 in Appendix A provides detailed variable definitions.

Variables	**Obs.**	**Mean**	**P25**	**Median**	**P75**	**STD**
I (LS)	36,179	0.306	0.000	0.000	1.000	0.461
#LS Sentences	36,179	0.877	0.000	0.000	1.000	2.492
Log(1+#LS Sentences)	36,179	0.345	0.000	0.000	0.693	0.610
$\frac{\#LS\ Sentences}{\#Labor\ Sentences}$	36,179	0.020	0.000	0.000	0.018	0.050
LS Exposure^{SW}	36,561	0.068	0.000	0.000	0.061	0.195
LS Exposure^{Positive}	36,179	0.010	0.000	0.000	0.000	0.039
LS Exposure^{NonPositive}	36,179	0.049	0.000	0.000	0.051	0.123
LS Exposure^{Mgmt}	36,179	0.023	0.000	0.000	0.000	0.077
LS Exposure$^{Q\&A}$	36,179	0.036	0.000	0.000	0.000	0.099
FF3F-adjusted Stock Return	34,878	0.003	-0.235	-0.038	0.163	0.450
FF5F-adjusted Stock Return	34,878	0.011	-0.234	-0.034	0.174	0.460
△CAPEX	33,784	0.165	-0.269	0.000	0.254	1.040
△R&D	32,578	0.0001	0.000	0.000	0.000	0.161
△CW Process Patent	32,991	0.280	0.000	0.000	0.000	11.367
OCF	36,109	0.048	0.016	0.067	0.122	0.164
Ind.adj OCF	36,109	0.002	-0.039	0.005	0.063	0.149
△AI Employee Share	26,151	0.0001	0.0000	0.0000	0.0000	0.0010
#CW AI Patents/AT	16,182	0.0011	0.0000	0.0000	0.0002	0.0074
MA3-△LS^{Increase}	31,883	0.015	0.000	0.000	0.002	0.041
MA3-△LS^{Decrease}	31,883	-0.010	0.000	0.000	0.000	0.033
Log(Per Employee Staff Expenses)	20,491	4.512	3.908	4.559	5.209	1.039
Employees/AT	32,023	0.004	0.001	0.002	0.004	0.006

Table A4. Keyword List from *Word2Vec*

Panel A. Labor-related keyword list

Seed words	labor, manpower, staff, personnel, people, worker, human, employee, workforce
Expanded words from Word2Vec	wage, hourly, salary, overhead, overtime, nurse, nursing, technician, headcount, payroll, salaried, part-time, furlough, therapist, engineer, sick, crew, rehire, team, member, frontline, absenteeism, full-time, training, pay, layoff, hire, hiring, workmen, job, roster, driver, contractor, skilled, work, trainee, leave, talent, blue-collar, head count, recruit

Panel B. Labor-shortage-related keyword list

Seed words	labor shortage, manpower shortage, worker shortage, staff shortage, labor constraint, labor crisis, labor scarcity, labor market constraint, understaffing
Expanded words from Word2Vec	tight labor market, labor availability, shortage labor, labor challenge, driver shortage, absenteeism, labor market shortage, talent crunch, labor availability issue, tightening labor market, worker availability, labor market challenge, labor bottleneck, shortage skilled, labor shortage issue, nursing shortage, truck driver shortage, tightness labor market, staffing challenge, labor availability challenge, staff burnout, shortage skilled labor, construction labor shortage, employee absenteeism, labor tightness, workforce availability, employee shortage, labor market pressure, workforce disruption, supply and labor constraint, workforce constraint, staffing shortage, driver challenge, skill shortage, aging workforce, talent shortage, salary inflation and labor shortage, labor capacity constraint, staffing issue, hiring challenge, immigration restriction, labor supply issue, staffing availability, recruiting labor, full-employment, contractor shortage, staffing inefficiency, driver staffing, labor market disruption, overtime issue, employment challenge, absentee rate, hiring driver, manpower availability, personnel challenge, labor slowdown, , labor market tighten, driver availability, crew shortage, nurse attrition, employee absence, staff availability, recruitment issue, hiring issue, labor shortfall, short-staffed, labor recruitment, tightening job market, manpower restriction

Table A5. Prediction Performance in Classifying Labor-Shortage-Related Sentences

This table presents the prediction performance in classifying labor-shortage-related sentences in the testing sample using the fine-tuned *FinBert*. The testing sample contains 500 sentences, of which 314 are non-labor-shortage-related (negative) and 186 are labor-shortage-related (positive). The 500 testing sentences are randomly selected from the full sample of 5,000 sentences and are manually labeled by the authors. For each sentence category, we compare three dimensions of prediction performance, which are precision, recall, and f1-score, respectively. For the total testing sentence sample, we also report the overall accuracy, macro average, and weighted average. The overall accuracy is measured as the number of correctly classified sentences divided by the total number of sentences in the testing sample. The macro average represents the unweighted mean value for each category and does not take label imbalance into account. The weighted average represents the weighted mean value for each category and take into account the label imbalance. The precision is calculated as *true positives/(true positives + false postives)*. The recall is calculated as *true positives/(true positives + false negatives)*. The f1-score represents a harmonic mean of the precision and recall, which is measured as *2 × (precision × recall)/(precision + recall)*.

	Precision	**Recall**	**F1-score**	**# Sentence**
Negative	0.97	0.95	0.96	314
Positive	0.91	0.95	0.93	186
Overall Accuracy			**0.95**	500
Macro Average	0.94	0.95	0.94	500
Weighted Average	0.95	0.95	0.95	500

Table A6. Labor-Shortage-Related Sample Sentences from Conference Call Transcripts

This table reports 20 randomly selected labor-shortage-related sentences that are predicted by the fine-tuned *FinBert* model.

Examples of Labor-shortage-related Sentence	Company	Year-Quarter
1. We are experiencing some inflationary pressures that are more annualized but – in our cost of equipment and cost of parts, and we're definitely going to see some cost pressure in compensation and labor as the market tightens, and the labor market especially is getting tightened at a few of the growth plays.	ARCHROCK INC	2017Q4
2. And the product – the lack of improved efficiency year-over-year in the factory, in large part, is driven by the labor scarcity.	LENNOX INTERNATIONAL INC	2019Q4
3. We've launched remediation plans to be back on track, but unfortunately, staffing issues limited some of the positive momentum of our results in this quarter.	SCHOOL SPECIALTY INC	2018Q3
4. Business leaders have also noted that one of their emerging concerns for the region is tightening labor supply company by upward pressure on wages.	COLUMBIA BANKING SYSTEM INC	2017Q2
5. The root cause is largely domestic, heavily tied to the labor availability staffing challenges that in all businesses are essentially facing today.	DEL TACO RESTAURANTS INC	2021Q3
6. Many of the productivity improvement initiatives and some of the near-term issues pass as a result of the productivity issues, like some of the labor constraint issues that we faced.	WABASH NATIONAL CORP	2017Q4
7. The Mayodan guys are running full out, a lot of overtime, and we're working hard to increase production there.	STURM RUGER & CO INC	2015Q3
8. The difficulty recruiting and retaining qualified drivers continues to be a challenge, and we have not been paid commensurately for the services we provide.	COVENANT LOGISTICS GROUP INC	2012Q2
9. I would like to have gotten there sooner, but I think we're finding the hiring environment is pretty intense out there.	ARIBA INC	2010Q3
10. And just last question on the labor piece, and I understand there's pressures now and you're anxious to get people hired.	DENNYS CORP	2021Q2
11. High U.S. employment levels created wage pressure and made it increasingly difficult to attract and retain employees.	BRINKS CO	2020Q1
12. The increase in compensation and benefits as a percentage of company-owned store sales reflected the increased shift of the portfolio to markets with lower sales volumes and continued labor inefficiencies associated with new store openings.	PAPA MURPHY'S HOLDINGS INC	2017Q3

13. And so we had a personnel issue where that was not getting done as timely as it was in the past and we made changes there, redirected resources and got that caught up.	AIR METHODS CORP	2012Q4
14. This catch up of new equipment, combined with the challenging driver recruiting market, were the key contributing factors to our unseated tractor percentage.	USA TRUCK INC	2019Q1
15. But as highlighted in last quarter's call, labor shortages are limiting growth.	BAKER HUGHES INC	2012Q1
16. As you navigate your way through that, we obviously will have a potential labor headwind.	JETBLUE AIRWAYS CORP	2017Q4
17. Well, I mean listen, labor was tight before we got into this pandemic.	OMEGA HEALTHCARE INVS INC	2020Q2
18. With the continued tightness in the labor markets, we are focused on investments in manufacturing automation, continuous improvement initiatives and Lean Six Sigma implementation throughout our organization.	CORNERSTNE BULDNG BRNDS INC	2019Q4
19. We currently expect year-over-year F&E revenue improvement versus this year's softness related to COVID absenteeism and inspection delays, which occurred primarily in the second and third fiscal quarters.	REV GROUP INC	2021Q1
20. The other issue that we definitely continue to face out on the job site – really plays into our strengths – is the labor shortage.	BUILDERS FIRSTSOURCE	2015Q3

Table A8. Correlation Matrix of Firm-Level Labor-Shortage Exposure

This table reports the correlation matrix of *LS Exposure* and its lags (panel A) and the correlation matrix of *I (LS)* and its lags (panel B). *LS Exposure* is a firm's labor-shortage exposure in year *t*. *I (LS)* is an indicator variable that equals one if the *LS Exposure* of a firm in that year is larger than zero, and equals zero otherwise.

Panel A. LS Exposure

VARIABLES	LS Exposure	LS Exposure $_{t-1}$	LS Exposure $_{t-2}$	LS Exposure $_{t-3}$	LS Exposure $_{t-4}$
LS Exposure	1.000				
LS Exposure $_{t-1}$	0.562	1.000			
LS Exposure $_{t-2}$	0.456	0.566	1.000		
LS Exposure $_{t-3}$	0.428	0.454	0.564	1.000	
LS Exposure $_{t-4}$	0.351	0.382	0.434	0.544	1.000

Panel B. LS Indicator

VARIABLES	**I (LS)**	I (LS) $_{t-1}$	I (LS) $_{t-2}$	I (LS) $_{t-3}$	I (LS) $_{t-4}$
I (LS)	1.000				
I (LS) $_{t-1}$	0.333	1.000			
I (LS) $_{t-2}$	0.283	0.326	1.000		
I (LS) $_{t-3}$	0.253	0.272	0.318	1.000	
I (LS) $_{t-4}$	0.226	0.238	0.255	0.301	1.000

Table A9. Firm Characteristics and Labor-Shortage Exposure

This table reports the regression results that investigate the relations between various firm characteristics and labor-shortage exposure. The dependent variable *LS Exposure* is a firm's labor-shortage exposure in year *t*. *I (LS)* is an indicator variable that equals one if the *LS Exposure* of a firm in that year is larger than zero, and equals zero otherwise. Columns 1-4 (5-8) investigate the relations between various firm characteristics and *LS Exposure* (*I (LS)*) of a firm in the same year. Columns 1 and 5 do not include any fixed effects. Columns 2 and 6 include year fixed effects. Columns 3 and 7 include both year fixed effects and industry fixed effects. Columns 4 and 8 include industry-by-year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
VARIABLES	LS Exposure				I (LS)			
ROA	-0.031***	0.000	0.003	-0.009	-0.117***	-0.047**	-0.057***	-0.079***
	(0.007)	(0.006)	(0.007)	(0.006)	(0.022)	(0.022)	(0.021)	(0.021)
Book Leverage	-0.005	-0.015***	-0.012***	-0.016***	-0.025**	-0.047***	-0.035***	-0.039***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.011)	(0.011)	(0.010)	(0.010)
Stock Return	0.013***	-0.003	-0.003	-0.006***	0.028***	-0.005	-0.005	-0.006
	(0.002)	(0.002)	(0.002)	(0.002)	(0.005)	(0.006)	(0.006)	(0.006)
CAPEX	-0.145***	-0.020	-0.031	-0.068*	-0.189**	0.157*	0.147*	-0.018
	(0.030)	(0.030)	(0.034)	(0.037)	(0.093)	(0.094)	(0.089)	(0.094)
MTB	0.000	-0.000	-0.000	-0.000	0.003***	0.002***	0.002***	0.002***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.000)	(0.000)
Sales Growth	0.014***	0.006**	0.008***	0.003	0.016**	0.009	0.018**	0.008
	(0.002)	(0.002)	(0.002)	(0.002)	(0.008)	(0.008)	(0.008)	(0.008)
Firm Size	0.004***	0.004***	0.003***	0.003***	0.032***	0.033***	0.030***	0.031***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
Cash	-0.015**	-0.025***	-0.029***	-0.021***	0.012	-0.019	-0.032	-0.020
	(0.007)	(0.007)	(0.008)	(0.008)	(0.022)	(0.022)	(0.022)	(0.022)
Asset Tangibility	0.062***	0.046***	0.030***	0.035***	0.167***	0.119***	0.084***	0.109***
	(0.009)	(0.009)	(0.011)	(0.011)	(0.024)	(0.024)	(0.027)	(0.028)
R&D	-0.113***	-0.080***	-0.053***	-0.067***	-0.402***	-0.310***	-0.216***	-0.237***
	(0.009)	(0.009)	(0.010)	(0.010)	(0.032)	(0.032)	(0.033)	(0.033)
Employees/AT	4.599***	5.062***	5.094***	6.139***	10.591***	11.729***	9.995***	11.185***
	(0.470)	(0.469)	(0.656)	(0.663)	(0.758)	(0.748)	(0.934)	(0.932)
Year FE	No	Yes	Yes	No	No	Yes	Yes	No
Industry FE	No	No	Yes	No	No	No	Yes	No
Industry-Year FE	No	No	No	Yes	No	No	No	Yes
Obs.	33,534	33,534	33,534	33,463	33,534	33,534	33,534	33,463
Adj. R2	0.058	0.110	0.184	0.253	0.067	0.098	0.150	0.165

Table A10. Bottom-10 Industries by Average Labor-Shortage Exposure

This table reports the bottom-10 industries by average labor-shortage exposure.

2-Digit SIC	Industry	LS Exposure
02	Agricultural Production - Livestock	0.000
08	Forestry	0.000
09	Fishing, Hunting and Trapping	0.000
76	Miscellaneous Repair Services	0.000
86	Membership Organizations	0.000
89	Miscellaneous Services	0.000
21	Tobacco Products	0.009
28	Chemicals and Allied Products	0.016
48	Communications	0.017
99	Non-classifiable Establishments	0.023

Table A11. Implications of Firm-Level Labor-Shortage Exposure: Fama-French Three (Five)-Factor-Adjusted Stock Returns

This table reports the regression results that investigate the implications of firm-level labor-shortage exposure on one-year-ahead cross-sectional stock returns. In Panel A, the dependent variable *FF3F-adjusted Stock Return* is the Fama-French three-factor-adjusted stock returns. In Panel B, the dependent variable *FF5F-adjusted Stock Return* is the Fama-French five-factor-adjusted stock returns. The independent variable *LS Exposure* is a firm's labor-shortage exposure in a year. For each stock, we use the past year's daily returns to estimate the stock's three-factor (five-factor) exposures by running time-series regressions. We next calculate the factor-adjusted daily returns over the next year using the estimated factor loadings and the realized factor returns (factor-adjusted daily returns are calculated as the difference between the realized excess returns of the stock and the expected excess returns from the Fama-French three-factor or five-factor model). We then compound the daily adjusted returns into annual adjusted returns. All specifications include firm fixed effects and year (or industry-by-year) fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. Fama-French three-factor-adjusted stock returns

	(1)	(2)	(3)	(4)
VARIABLES		FF3F-adjusted Stock Return $_{t+1}$		
LS Exposure	**-0.071*** **	**-0.068*** **	**-0.060*** **	**-0.056*** **
	(0.020)	**(0.020)**	**(0.021)**	**(0.021)**
Firm Size		-0.105***		-0.100***
		(0.009)		(0.009)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Industry-Year FE	No	No	Yes	Yes
Obs.	34,708	34,708	34,706	34,706
Adj. R2	0.057	0.069	0.113	0.123

Panel B. Fama-French five-factor-adjusted stock returns

	(1)	(2)	(3)	(4)
VARIABLES		FF5F-adjusted Stock Return $_{t+1}$		
LS Exposure	**-0.061*** **	**-0.058*** **	**-0.048** **	**-0.045** **
	(0.020)	**(0.020)**	**(0.021)**	**(0.021)**
Firm Size		-0.105***		-0.100***
		(0.009)		(0.010)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Industry-Year FE	No	No	Yes	Yes
Obs.	34,708	34,708	34,706	34,706
Adj. R2	0.057	0.068	0.110	0.119

Table A12. Implications of Firm-Level Labor-Shortage Exposure: Operating Cash Flow

This table reports the regression results that investigate the implications of firm-level labor-shortage exposure on one-year-ahead operating cash flow. The dependent variable *Ind-adj.OCF* is measured as a firm's operating cash flow divided by the firm's book value of total assets, further adjusted by the average *OCF* of firms in the same industry (2-digit SIC) and year. *OCF* is measured as a firm's operating cash flow divided by the firm's book value of total assets. The independent variable *LS Exposure* is a firm's labor-shortage exposure in a year. Columns 1-2 (3-4) include firm and year (industry-year) fixed effects, and we further control for firm size in Columns 2 and 4. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES	Ind-adj. OCF $_{t+1}$		OCF $_{t+1}$	
LS Exposure	**-0.011*** **	**-0.012*** **	**-0.005**	**-0.006**
	(0.004)	**(0.004)**	**(0.004)**	**(0.004)**
Firm Size		0.035***		0.032***
		(0.003)		(0.003)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Industry-Year FE	No	No	Yes	Yes
Obs.	36,109	36,109	36,109	36,109
Adj. R2	0.621	0.632	0.694	0.701

Table A13. Industry Labor-Shortage Exposure and Firms' Operating Performance and Policy Responses

The table compares stock and operating performance and firm policy responses based on their industry-year-level labor-shortage exposure. We construct an indicator, *I* ($LS^{ind\text{-}top20\%}$), that equals one if a firm's industry (2-digit SIC) falls within the top quintile of industry-average *LS Exposure* across all industries in a given year and zero otherwise. All specifications include control variables, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

VARIABLES	(1) Stock Return $_{t+1}$	(2) SIC3-adj.ROA $_{t+1}$	(3) CAPEX $_{t+1}$	(4) R&D $_{t+1}$	(5) △Employees/AT $_{t+1}$	(6) Process Claims Share $_{t+1,t+3}$	(7) #CW Process Patents/AT $_{t+1,t+3}$
I ($LS^{ind\text{-}top20\%}$)	-0.059***	-0.003	0.004***	0.007***	0.000	0.018***	0.004***
	(0.011)	(0.002)	(0.001)	(0.001)	(0.000)	(0.005)	(0.001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	36,179	36,109	31,926	20,025	32,023	16,466	16,466
Adj. R2	0.227	0.660	0.729	0.665	0.049	0.677	0.591

Table A14. Alternative Measure of Persistent Labor-Shortage Exposure

The table reports the regression results that investigate the heterogeneous effects of firm-level labor-shortage exposure based on alternative measure of persistent labor-shortage exposure. The indicator variable, *Repeated LS (1/0)*, equals one if a firm has experienced labor shortages at least once during the five years prior to the current LS exposure, otherwise equals zero. In all specifications, we include control variables and firm and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

VARIABLES	(1) Growth in Per-Emp. Staff Exp.$_{t+1}$	(2) Stock Return$_{t+1}$	(3) Ind-adj.ROA$_{t+1}$	(4) CAPEX$_{t+1}$	(5) R&D$_{t+1}$	(6) $\triangle$Employees/AT$_{t+1}$	(7) Process Claims Share$_{t+1,t+3}$	(8) CW Process Patents/AT$_{t+1,t+3}$
LS Exposure	0.041	-0.078*	0.000	0.011***	-0.004	0.0000	-0.006	-0.011**
	(0.033)	(0.040)	(0.010)	(0.003)	(0.008)	(0.0001)	(0.019)	(0.005)
Repeated LS (1/0)	0.005	-0.011	-0.004**	-0.000	0.003	-0.0000	0.009*	-0.005
	(0.004)	(0.008)	(0.002)	(0.001)	(0.002)	(0.0000)	(0.005)	(0.004)
LS Exposure × Repeated LS (1/0)	**-0.008**	**-0.011**	**-0.019***	**-0.004**	**0.027*****	**-0.0003***	**0.082*****	**0.025*****
	(0.037)	**(0.043)**	**(0.011)**	**(0.003)**	**(0.008)**	**(0.0002)**	**(0.021)**	**(0.008)**
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	18,631	35,234	35,168	31,108	19,519	31,200	16,442	16,442
Adj. R2	0.032	0.226	0.689	0.731	0.665	0.053	0.678	0.606

Table A15. Alternative Measures of Labor-Shortage Exposure

The table examines the robustness of the baseline results using alternative measures of firm-level labor-shortage exposure. In Panel A, we use the average raw number of labor-shortage sentences across conference calls in a given firm-year, *#LS Sentences*, without normalizing by total sentence count. In Panel B, we further take a natural logarithm of average raw number of labor-shortage sentences: *Log(1+#LS Sentences)*, to reduce skewness and account for nonlinearities. In Panel C, we generate $\frac{\#LS\ Sentences}{\#Labor\ Sentences}$, which divided the raw number of labor-shortage sentences by the number of labor-related sentences to control for variation in labor discussion intensity. In Panel D, we further consider the severity of labor-shortage concerns. We assign each labor-shortage sentence a weight equal to (number of severity-related words×0.5)+1. $LS\ Exposure^{SW}$ is then the severity-weighted labor-shortage exposure, measured as the severity-weighted sentence count scaled by total sentence count and averaged across all conference calls in the year. In Panel E, we classify labor-shortage sentences by sentiment using *FinBert*, and construct separate $LS\ Exposure^{Positive}$ and $LS\ Exposure^{NonPositive}$ measures. All specifications include control variables, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. Raw count of labor-shortage sentences

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
VARIABLES	Stock Return $_{t+1}$	Ind-adj.ROA $_{t+1}$	CAPEX $_{t+1}$	R&D $_{t+1}$	△Employees/AT $_{t+1}$	Process Claims Share $_{t+1,t+3}$	#CW Process Patents/AT $_{t+1,t+3}$
#LS Sentences	-0.006***	-0.001***	0.0004***	0.001***	-0.00001**	0.004***	0.0005***
	(0.001)	(0.000)	(0.0001)	(0.000)	(0.00000)	(0.001)	(0.0001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	36,179	36,109	31,926	20,025	32,023	16,466	16,466
Adj. R2	0.226	0.688	0.729	0.665	0.050	0.677	0.591

Panel B. Log-transformed count of labor-shortage sentences

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
VARIABLES	Stock Return $_{t+1}$	Ind-adj.ROA $_{t+1}$	CAPEX $_{t+1}$	R&D $_{t+1}$	△Employees/AT $_{t+1}$	Process Claims Share $_{t+1,t+3}$	#CW Process Patents/AT $_{t+1,t+3}$
Log(1+#LS Sentences)	-0.033***	-0.005***	0.002***	0.005***	-0.00003*	0.012***	0.001***
	(0.005)	(0.001)	(0.000)	(0.001)	(0.00002)	(0.003)	(0.001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	36,179	36,109	31,926	20,025	32,023	16,466	16,466
Adj. R2	0.227	0.688	0.729	0.665	0.049	0.677	0.591

Panel C. Labor-shortage sentences normalized by labor-related sentences

VARIABLES	(1) Stock Return $_{t+1}$	(2) Ind-adj.ROA $_{t+1}$	(3) CAPEX $_{t+1}$	(4) R&D $_{t+1}$	(5) △Employees/AT $_{t+1}$	(6) Process Claims Share $_{t+1,t+3}$	(7) #CW Process Patents/AT $_{t+1,t+3}$
$\frac{\#LS\ Sentences}{\#Labor\ Sentences}$	-0.211***	-0.044***	0.019***	0.022***	-0.0001	0.066**	0.012***
	(0.049)	(0.011)	(0.005)	(0.007)	(0.0001)	(0.028)	(0.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	36,179	36,109	31,926	20,025	32,023	16,466	16,466
Adj. R2	0.226	0.688	0.729	0.665	0.049	0.676	0.591

Panel D. Severity weighted labor-shortage exposure

VARIABLES	(1) Stock Return $_{t+1}$	(2) Ind-adj.ROA $_{t+1}$	(3) CAPEX $_{t+1}$	(4) R&D $_{t+1}$	(5) △Employees/AT $_{t+1}$	(6) Process Claims Share $_{t+1,t+3}$	(7) #CW Process Patents/AT $_{t+1,t+3}$
LS Exposure^{SW}	-0.062***	-0.014***	0.005***	0.014***	-0.0003***	0.044***	0.006***
	(0.014)	(0.003)	(0.001)	(0.003)	(0.0001)	(0.009)	(0.002)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	36,179	36,109	31,926	20,025	32,023	16,466	16,466
Adj. R2	0.226	0.688	0.729	0.665	0.050	0.677	0.591

Panel E. Labor-shortage exposure by sentiment

VARIABLES	(1) Stock Return $_{t+1}$	(2) Ind-adj.ROA $_{t+1}$	(3) CAPEX $_{t+1}$	(4) R&D $_{t+1}$	(5) △Employees/AT $_{t+1}$	(6) Process Claims Share $_{t+1,t+3}$	(7) #CW Process Patents/AT $_{t+1,t+3}$
LS Exposure^{Positive}	-0.034	-0.020	0.007	0.006	-0.0003	0.126***	0.019
	(0.069)	(0.013)	(0.006)	(0.014)	(0.0003)	(0.038)	(0.012)
LS Exposure^{NonPositive}	-0.108***	-0.020***	0.008***	0.024***	-0.0003**	0.054***	0.007***
	(0.025)	(0.005)	(0.002)	(0.004)	(0.0001)	(0.013)	(0.003)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	36,179	36,109	31,926	20,025	32,023	16,466	16,466
Adj. R2	0.226	0.688	0.729	0.665	0.050	0.677	0.591

Table A16. Firm-Level Labor-Shortage Exposure, Geographic Dispersion, and Future Stock Returns and Operating Performance

This table reports the regression results that investigate whether geographic dispersion can help mitigate the negative effects of firm-level labor-shortage exposure on future stock returns and operating performance. We measure a firm's one-year-ahead stock returns using raw stock returns (*Stock Return*), Fama-French three-factor-adjusted stock returns (*FF3F-adjusted Stock Return*), or Fama-French five-factor-adjusted stock returns (*FF5F-adjusted Stock Return*). We measure a firm's one-year-ahead operating performance using *Ind-adj.ROA*. *LS Exposure* is a firm's labor-shortage exposure in a year. In Panel A, *GeoDis* is measured as the natural logarithm of the number of unique counties in which a firm's establishments (obtained from the NETS database) operated in a year. In Panel B, $GeoDis^{DR}$ is measured as the decile rank of the number of unique counties in which a firm's establishments operated in a year. All specifications include control variables, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. The role of geographic dispersion (raw value)

VARIABLES	(1) Stock Return $_{t+1}$	(2) FF3F-adj. Stock Return $_{t+1}$	(3) FF5F-adj. Stock Return $_{t+1}$	(4) Ind-adj.ROA $_{t+1}$
LS Exposure	-0.123**	-0.184***	-0.176***	-0.032**
	(0.058)	(0.066)	(0.066)	(0.015)
GeoDis	0.012	-0.002	-0.006	-0.007
	(0.014)	(0.016)	(0.016)	(0.004)
LS Exposure×GeoDis	**0.009**	**0.034****	**0.032***	**0.006**
	(0.016)	**(0.016)**	**(0.017)**	**(0.004)**
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs.	23,457	22,710	22,710	23,415
Adj. R2	0.217	0.040	0.040	0.693

Panel B. The role of geographic dispersion (decile rank)

VARIABLES	(1) Stock Return $_{t+1}$	(2) FF3F-adj. Stock Return $_{t+1}$	(3) FF5F-adj. Stock Return $_{t+1}$	(5) Ind-adj.ROA $_{t+1}$
LS Exposure	-0.141**	-0.225***	-0.210**	-0.045**
	(0.071)	(0.084)	(0.085)	(0.019)
$GeoDis^{DR}$	0.004	-0.001	-0.001	0.001
	(0.006)	(0.007)	(0.007)	(0.002)
LS Exposure×$GeoDis^{DR}$	**0.006**	**0.022****	**0.020***	**0.005***
	(0.009)	**(0.010)**	**(0.011)**	**(0.002)**
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs.	23,457	22,710	22,710	23,415
Adj. R2	0.217	0.040	0.040	0.693

Table A17. Corporate Policy Responses to Labor-Shortage Exposure and Future Stock Returns and Operating Performance

This table reports the regression results that investigate whether corporate policy responses to labor-shortage exposure help improve future stock and operating performance. The dependent variable *Stock Return* is a firm's buy-and-hold stock return; *Ind-adj.ROA* is measured as a firm's operating income before depreciation divided by the firm's book value of total assets, further adjusted by the average *ROA* of firms in the same industry (2-digit SIC) and year. The independent variable *LS Exposure* is a firm's labor-shortage exposure in a year; $\triangle$*CAPEX* is measured as the change in a firm's capital expenditure from year *t-1* to year *t*, divided by the firm's capital expenditure in year *t-1*; $\triangle$*R&D* is measured as the change in a firm's R&D expenses from year *t-2* to year *t-1*, divided by the firm's R&D expenses in year *t-2*; $\triangle$*CW Process Patent* is measured as the change in a firm's number of citation-weighted process patents from year *t-2* to year *t-1*, divided by the firm's number of citation-weighted process patents in year *t-2*. All specifications control for firm size, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	Stock Return $_{t+1}$			Ind-adj.ROA $_{t+1}$		
LS Exposure	-0.097***	-0.090***	-0.164***	-0.015***	-0.013***	-0.009
	(0.021)	(0.020)	(0.032)	(0.004)	(0.004)	(0.007)
$\triangle$CAPEX	-0.013***			-0.000		
	(0.003)			(0.001)		
$\triangle$R&D $_{t-1}$		0.061***			-0.022***	
		(0.020)			(0.006)	
$\triangle$CW Process Patent $_{t-1}$			-0.001***			-0.000**
			(0.000)			(0.000)
LS Exposure×$\triangle$CAPEX	**0.039****			**0.002**		
	(0.016)			**(0.004)**		
LS Exposure×$\triangle$R&D $_{t-1}$		**0.051**			**0.029**	
		(0.111)			**(0.023)**	
LS Exposure×$\triangle$CW Process Patent $_{t-1}$			**0.011*****			**0.002*****
			(0.003)			**(0.001)**
Firm Size	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	33,599	32,360	15,139	33,534	32,926	15,114
Adj. R2	0.225	0.225	0.204	0.688	0.685	0.731

Table A18. Corporate Policy Responses and Future Labor-Shortage Exposure

This table reports the regression results that investigate whether corporate policy responses help reduce the a firm's exposure to labor shortages in the future. The dependent variable *LS Exposure* is the labor-shortage exposure of a firm in a year. The independent variable, $\triangle CAPEX$, is the change in a firm's capital expenditure from year *t-1* to year *t*, divided by the firm's capital expenditure in year *t-1*; $\triangle R\&D$ is the change in a firm's R&D expenses from year *t-1* to year *t*, divided by the firm's R&D expenses in year *t-1*; $\triangle CW\ Process\ Patent$ is the change in a firm's number of citation-weighted process patents from year *t-1* to year *t*, divided by the firm's number of citation-weighted process patents in year *t-1*. All specifications control for firm size, firm fixed effects, and year (industry-year) fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
VARIABLES				LS Exposure $_{t+1}$				
△CAPEX	-0.002***	-0.002***					-0.002***	-0.002***
	(0.001)	(0.001)					(0.001)	(0.001)
△R&D			0.003	0.002			0.003	0.002
			(0.005)	(0.005)			(0.005)	(0.005)
△CW Process Patent					-0.000**	-0.000**	-0.000	-0.000
					(0.000)	(0.000)	(0.000)	(0.000)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No	Yes	No
Industry-Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Obs.	32,779	32,706	30,097	30,015	34,566	34,489	29,801	29,719
Adj. R2	0.376	0.433	0.378	0.436	0.375	0.432	0.379	0.436

Table A19. Extensive Margin vs. Intensive Margin

The table compares the extensive margin and intensive margin of the effects of firm-level labor-shortage exposure on stock price reactions, future stock returns and operating performance, and corporate policy responses. We conduct the extensive margin analyses by replacing the continuous labor-shortage exposure measure with an indicator *I (LS)* that equals one if *LS Exposure* is larger than zero, and equals zero otherwise. We conduct the intensive margin analyses by restricting to the sample of firms that are exposed to labor shortages (i.e., *LS Exposure* is larger than zero in a firm-year). Panel A (B) examines the extensive (intensive) margin of the effects of firm-level labor-shortage exposure on stock price reactions and future stock returns and operating performance. Panel C (D) examines the extensive (intensive) margin of the effects of firm-level labor-shortage exposure on corporate policy responses. All specifications include control variables, firm fixed effects, and year (year-quarter) fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. Extensive Margin Analysis: Stock price reactions to labor-shortage exposure and implications on future stock returns and operating performance

VARIABLES	(1) CAR(0,2)	(2) Stock Return $_{t+1}$	(3) Ind-adj.ROA $_{t+1}$
I (LS)	**-0.005*** **	**-0.031*** **	**-0.004*** **
	(0.001)	**(0.006)**	**(0.001)**
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year (Year-Quarter) FE	Yes	Yes	Yes
Obs.	119,004	36,179	36,109
Adj. R2	0.028	0.226	0.688

Panel B. Intensive Margin Analysis: Stock price reactions to labor-shortage exposure and implications on future stock returns and operating performance

VARIABLES	(1) CAR(0,2)	(2) Stock Return $_{t+1}$	(3) Ind-adj.ROA $_{t+1}$
LS Exposure	**-0.006***	**-0.090*** **	**-0.004**
	(0.003)	**(0.025)**	**(0.005)**
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year (Year-Quarter) FE	Yes	Yes	Yes
Obs.	15,555	10,173	10,167
Adj. R2	0.035	0.256	0.671

Panel C. Extensive Margin Analysis: Corporate policy responses to labor-shortage exposure

VARIABLES	(1) CAPEX $_{t+1}$	(2) R&D $_{t+1}$	(3) △Employees/AT $_{t+1}$	(4) Process Claims Share $_{t+1,\ t+3}$	(5) # CW Process Patents/AT $_{t+1,t+3}$
I (LS)	**0.002*** **	**0.003*** **	**-0.000**	**0.007** **	**0.000**
	(0.000)	**(0.001)**	**(0.000)**	**(0.003)**	**(0.001)**
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Obs.	31,926	20,025	32,023	16,466	16,466
Adj. R2	0.729	0.665	0.049	0.676	0.591

Panel D. Intensive Margin Analysis: Corporate policy responses to labor-shortage exposure

VARIABLES	(1) CAPEX $_{t+1}$	(2) R&D $_{t+1}$	(3) △Employees/AT $_{t+1}$	(4) Process Claims Share $_{t+1,\ t+3}$	(5) # CW Process Patents/AT $_{t+1,t+3}$
LS Exposure	**0.006*** **	**0.010*** **	**-0.0003*** **	**0.034*** **	**0.007**
	(0.002)	**(0.003)**	**(0.0001)**	**(0.012)**	**(0.005)**
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Obs.	8,705	5,101	8,723	4,239	4,239
Adj. R2	0.729	0.752	0.061	0.687	0.458

Table A20. Management Presentation Section vs. Q&A Section

The table examines the robustness of the results using firm-level labor-shortage exposure constructed from the management presentation section or the Q&A section of the earnings conference call transcripts. Panel A investigates the stock price reactions to firm-level labor-shortage exposure and the implications of firm-level labor-shortage exposure on future stock returns and operating performance. Panel B examines the corporate policy responses to firm-level labor-shortage exposure. *LS Exposure*^{Mgmt} is a firm's labor-shortage exposure measured using the management presentation section of the earnings conference call transcripts in a year (or year-quarter). *LS Exposure*$^{Q\&A}$ is a firm's labor-shortage exposure measured using the Q&A section of the earnings conference call transcripts in a year (or year-quarter). All specifications include control variables, firm fixed effects, and year (year-quarter) fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. Stock price reactions to labor-shortage exposure and implications on future stock returns and operating performance

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES		CAR(0,2)			Stock Return $_{t+1}$			Ind-adj.ROA $_{t+1}$	
LS Exposure^{Mgmt}	-0.020***		-0.019***	-0.124***		-0.084**	-0.040***		-0.038***
	(0.003)		(0.003)	(0.038)		(0.041)	(0.007)		(0.008)
LS Exposure$^{Q\&A}$		-0.007***	-0.003*		-0.115***	-0.092***		-0.015***	-0.005
		(0.002)	(0.002)		(0.026)	(0.027)		(0.006)	(0.006)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year (Year-Quarter) FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	119,004	119,004	119,004	36,179	36,179	36,179	36,109	36,109	36,109
Adj. R2	0.028	0.028	0.028	0.226	0.226	0.226	0.688	0.688	0.688

Panel B. Corporate policy responses to labor-shortage exposure with firm and year fixed effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
VARIABLES		CAPEX$_{t+1}$			R&D$_{t+1}$			△Employees/AT$_{t+1}$			Process Claims Share$_{t+1,t+3}$			#CW Process Patents/AT$_{t+1,t+3}$	
LS Exposure^{Mgmt}	0.006**		0.002	0.031***		0.024***	-0.0005***		-0.0004*	0.112***		0.093***	0.016***		0.013**
	(0.003)		(0.003)	(0.008)		(0.008)	(0.0002)		(0.0002)	(0.020)		(0.020)	(0.005)		(0.006)
LS Exposure$^{Q\&A}$		0.014***	0.013***		0.024***	0.019***		-0.0004***	-0.0003**		0.074***	0.052***		0.010***	0.007*
		(0.003)	(0.003)		(0.004)	(0.005)		(0.0002)	(0.0001)		(0.017)	(0.016)		(0.004)	(0.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	31,926	31,926	31,926	20,025	20,025	20,025	32,023	32,023	32,023	16,466	16,466	16,466	16,466	16,466	16,466
Adj. R2	0.729	0.729	0.729	0.665	0.665	0.665	0.050	0.050	0.050	0.677	0.677	0.677	0.591	0.591	0.591

Table A21. Controlling for CEO Fixed Effects

This table examines the robustness of the results after controlling for CEO fixed effects, where CEO identities are obtained from earnings conference call transcripts. Panel A investigates the stock price reactions to firm-level labor-shortage exposure and the implications of firm-level labor-shortage exposure on future stock returns and operating performance. Panel B examines the corporate policy responses to firm-level labor-shortage exposure. All specifications include control variables, CEO fixed effects, firm fixed effects, and year (year-quarter) fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. Stock price reactions to labor-shortage exposure and implications on future stock returns and operating performance

VARIABLES	(1) CAR(0,2)	(2) Stock Return $_{t+1}$	(3) Ind-adj.ROA $_{t+1}$
LS Exposure	-0.012***	-0.081***	-0.011**
	(0.002)	(0.026)	(0.005)
Controls	Yes	Yes	Yes
CEO FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year (Year-Quarter) FE	Yes	Yes	Yes
Obs.	115,342	33,934	33,867
Adj. R2	0.022	0.168	0.711

Panel B. Corporate policy responses to labor-shortage exposure

VARIABLES	(1) CAPEX $_{t+1}$	(2) R&D $_{t+1}$	(3) $\triangle$Employees/AT $_{t+1}$	(4) Process Claims Share $_{t+1,t+3}$	(5) #CW Process Patents/AT $_{t+1,t+3}$
LS Exposure	0.007***	0.013***	-0.0002*	0.035***	0.008*
	(0.002)	(0.004)	(0.0001)	(0.012)	(0.004)
Controls	Yes	Yes	Yes	Yes	Yes
CEO FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Obs.	29,850	18,687	29,944	15,302	15,302
Adj. R2	0.729	0.668	-0.072	0.746	0.566

Table A22. AI Investment

This table reports the regression results that investigate the implications of firm-level labor-shortage exposure on AI investment. In Panel A, we investigate the effects of labor shortage exposure on firms' AI investment. In Panel B, we split the sample into 2005-2012 and 2013-2021 and compare the heterogeneous effects of labor shortage exposure on AI investment between these two sample periods. In Panel C, we further examine the heterogeneous effects of labor shortage exposure on AI investment based on industry-level knowledge intensity, proxied by SG&A expenditures, R&D expenses, and patenting outputs. The dependent variable *△AI Employee Share* is measured as the change in a firm's AI employee share from year *t* to year *t+1*. *#CW AI Patents/AT* is measured as the citation-weighted number of AI-related patents a firm has applied (and later granted) scaled by the total assets of the firm. The independent variable *LS Exposure* is a firm's labor-shortage exposure in a year. All specifications include control variables, firm fixed effects, and year (or industry-by-year) fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. Baseline regressions

	(1)	(2)	(3)	(4)
VARIABLES	△AI Employee Share $_{t+1}$		#CW AI Patents/AT $_{t+1,t+3}$	
LS Exposure	0.0002***	0.0002***	0.0004***	0.0002
	(0.0000)	(0.0000)	(0.0002)	(0.0002)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
Industry-Year FE	No	Yes	No	Yes
Obs.	26,346	26,281	16,466	16,466
Adj. R2	0.024	0.018	0.649	0.636

Panel B. Earlier versus later years

	(1)	(2)	(3)	(4)
VARIABLES	△AI Employee Share $_{t+1}$		#CW AI Patents/AT $_{t+1,\ t+3}$	
Sample Period	2005-2012	2013-2021	2005-2012	2013-2021
LS Exposure	0.0000	0.0002***	-0.0000	0.0005*
	(0.0000)	(0.0001)	(0.0001)	(0.0003)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs.	11,144	14,684	7,375	8,967
Adj. R2	0.107	-0.010	0.784	0.683

Panel C. Knowledge intensity

	(1)	(2)	(3)	(4)
VARIABLES	△AI Employee Share $_{t+1}$		#CW AI Patents/AT $_{t+1,\ t+3}$	
	Low	High	Low	High
		Industry SG&A		
LS Exposure	0.0001*	0.0003***	0.0000	0.0007**
	(0.0000)	(0.0001)	(0.0002)	(0.0003)
Obs.	13,306	12,893	6,794	9,584
Adj. R2	-0.012	0.038	0.620	0.653
		Industry R&D		
LS Exposure	0.0001**	0.0002***	-0.0001	0.0006**
	(0.0000)	(0.0001)	(0.0002)	(0.0002)
Obs.	12,931	13,294	5,150	11,256
Adj. R2	-0.017	0.031	0.430	0.648
		Industry Patent Outputs		
LS Exposure	0.0001***	0.0002**	-0.0000	0.0005**
	(0.0000)	(0.0001)	(0.0001)	(0.0002)
Obs.	13,089	13,142	5,055	11,326
Adj. R2	-0.048	0.034	0.477	0.650
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Table A23. Non-Linear Effects of Labor-Shortage Exposure

The table examines the non-linear effects of firm-level labor-shortage exposure. Specifically, we define three mutually exclusive groups. Group 1 (reference) refers to the firms with zero LS Exposure. Group 2 is those firms with non-zero LS Exposure below the median of the year, while Group 3 is the firms with non-zero LS Exposure above the median of the year. We then include indicator variables for Groups 2 ($LS\ Exposure^{Group2}$) and Group 3 ($LS\ Exposure^{Group3}$) in the regressions. All specifications include control variables, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

VARIABLES	(1) Stock Return $_{t+1}$	(2) Ind-adj.ROA $_{t+1}$	(3) CAPEX $_{t+1}$	(4) R&D $_{t+1}$	(5) △Employees/AT $_{t+1}$	(6) Process Claims Share $_{t+1,t+3}$	(7) #CW Process Patents/AT $_{t+1,t+3}$
LS Exposure^{Group2}	-0.027***	-0.004***	0.001***	0.001	0.000	0.004	-0.001
	(0.007)	(0.001)	(0.000)	(0.001)	(0.000)	(0.003)	(0.002)
LS Exposure^{Group3}	-0.037***	-0.005***	0.004***	0.006***	-0.000	0.010***	0.002**
	(0.008)	(0.002)	(0.001)	(0.001)	(0.000)	(0.004)	(0.001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	36,179	36,109	31,926	20,025	32,023	16,466	16,466
Adj. R2	0.226	0.688	0.729	0.665	0.049	0.676	0.591

Table A24. The Effects of Positive and Negative Labor-Shortage Shocks

The table examines the effects of positive and negative labor-shortage shocks on firms' stock and operating performance and policy responses. Specifically, we construct a 3-year moving average of LS Exposure and compute the change between year t and t–1. We then define $MA3\text{-}\triangle LS^{Increase}$ ($MA3\text{-}\triangle LS^{Decrease}$) which equals the change when it is positive (negative), and zero otherwise. All specifications include control variables, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

VARIABLES	(1) Stock Return $_{t+1}$	(2) Ind-adj.ROA $_{t+1}$	(3) CAPEX $_{t+1}$	(4) R&D $_{t+1}$	(5) $\triangle$Employees/AT $_{t+1}$	(6) Process Claims Share $_{t+1,t+3}$	(7) #CW Process Patents/AT $_{t+1,t+3}$
MA3-$\triangle$LS^{Increase}	-0.262***	-0.036**	0.019***	0.042***	-0.000	0.129***	0.026***
	(0.070)	(0.014)	(0.005)	(0.010)	(0.000)	(0.038)	(0.009)
MA3-$\triangle$LS^{Decrease}	0.036	0.032	0.016**	-0.035*	0.000	-0.014	-0.001
	(0.096)	(0.021)	(0.007)	(0.019)	(0.000)	(0.041)	(0.015)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	31,479	31,432	27,645	17,420	27,723	14,741	14,741
Adj. R2	0.220	0.694	0.742	0.672	0.049	0.683	0.628

Table A25. Long-Term Effects on *CAPEX* and *R&D*

This table reports the regression results that investigate the long-term effects of firms' labor-shortage exposure on *CAPEX* and *R&D* expenses. In Panel A, we investigate the effects of labor shortage exposure on firms' *CAPEX* and *R&D* investments over the years *t+1* to *t+3*. In Panel B, we examine the role of persistent labor shortages in explaining the effects of labor-shortage exposure on long-term *CAPEX* and *R&D* investments. The dependent variable *CAPEX* is measured as a firm's capital expenditures divided by its total value of assets. *R&D* is measured as a firm's research and development expenses divided by its total value of assets. The independent variable *LS Exposure* is a firm's labor-shortage exposure in a year. All specifications include control variables, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. Long-term effects on CAPEX and R&Ds

VARIABLES	(1) CAPEX t+1	(2) CAPEX t+1,t+2	(3) CAPEX t+1,t+3	(4) R&D t+1	(5) R&D t+1,t+2	(6) R&D t+1,t+3
LS Exposure	0.008***	0.007***	0.004	0.018***	0.015***	0.022***
	(0.002)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	31,926	31,955	31,963	20,025	20,025	20,025
Adj. R2	0.729	0.634	0.636	0.665	0.556	0.560

Panel B. Persistent labor shortages and CAPEX and R&Ds

VARIABLES	(1) CAPEX t+1	(2) CAPEX t+1,t+2	(3) CAPEX t+1,t+3	(4) R&D t+1	(5) R&D t+1,t+2	(6) R&D t+1,t+3
LS Exposure	0.010***	0.009**	0.008**	0.003	-0.001	-0.003
	(0.003)	(0.004)	(0.004)	(0.007)	(0.007)	(0.008)
Repeated LS (1/0)	0.000	-0.001	-0.001	0.004***	0.001	0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
LS Exposure × Repeated LS (1/0)	**-0.002**	**-0.003**	**-0.006**	**0.021*** **	**0.022*** **	**0.035*** **
	(0.003)	**(0.005)**	**(0.005)**	**(0.007)**	**(0.007)**	**(0.008)**
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	31,088	31,114	31,121	19,503	19,503	19,503
Adj. R2	0.731	0.637	0.645	0.665	0.549	0.553

Table A26. Levels of Per-Employee Staff Expenses and Employee Investment

This table reports the regression results that investigate the effects of firms' labor-shortage exposure on the levels of per-employee staff expenses and employee investment. The dependent variable *Log(Per Employee Staff Expenses)* is measured as the natural logarithm of a firm's per-employee staff expense in a year. *Employees/AT* is measured as a firm's number of employees (in thousand) divided by the book value of asset in a year. The independent variable *LS Exposure* is a firm's labor-shortage exposure in a year. All specifications include control variables, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

	(1)	(2)
VARIABLES	Log(Per Employee Staff Expenses) $_{t+1}$	Employees/AT $_{t+1}$
LS Exposure	0.023	-0.0006***
	(0.025)	(0.0002)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Obs.	20,491	32,023
Adj. R2	0.922	0.920

Appendix B Additional Validation Tests

In Appendix B, we provide two additional validation tests for the firm-level labor-shortage exposure. The first test utilises the 2017 Trump Immigration Policy Reforms as a negative shock to the U.S. labor supply that could intensify the labor-shortage exposure of firms. The second test leverages the COVID-era Federal Pandemic Unemployment Compensation (FPUC) program introduced under the CARES Act as a plausibly exogenous labor supply shock to low-wage industrial firms, which potentially increases their exposures to labor shortages.

B1.1 U.S. Immigration Policy Reforms

In this section, we exploit the immigration policy reforms proposed by the former U.S. president Donald Trump in 2017 as a quasi-natural experiment to examine the validity of the firm-level measure of labor-shortage exposure.

After his inauguration in 2017, Donald Trump actively reformed the immigration policies in the U.S. following his campaign slogan "Buy American, Hire American", aiming at enhancing the restrictions on immigrants and protecting American workers (**?**). A series of executive orders and actions were implemented by the Trump administration, which include but are not limited to i) enhancing immigration enforcements and border security by building a US-Mexico border wall and increasing the construction of detention facilities to stem the flow of illegal entrants; ii) suspending refugee admissions from certain Muslim-majority countries (i.e., Iran, Iraq, Sudan, Syria, Libya, Somalia, and Yemen) and reducing the number of refugees to be admitted to the U.S. from 110,000 to 50,000 in FY2017; iii) increasing vetting and processing time for legal immigration; iv) tightening H-1B visa approvals to high-skilled foreign labors; and v) limiting family-based immigration (or chain migration) to those who are immediate family members (spouses and minor children) of the U.S citizens or green card holders. In essence, these immigration policy reforms seek to protect domestic workers by reducing the number of legal or illegal immigrants.

However, immigrants are an essential component of the U.S. labor force. According to the U.S. Bureau of Labor Statistics (BLS), there are 27 million foreign-born workers in 2016, which account for 16.9 percentage points of total labor force.[1] Therefore, the significantly tightened immigration policies reduced total foreign labor supply in the U.S. We thus use the 2017 immigration policy reforms as a quasi-natural experiment to validate the firm-level measure of labor-shortage exposure. We conjecture that the immigration policy reforms should lead to greater firm-level labor-shortage exposure for labor-intensive firms relative to other firms.

We partition our sample firms into labor-intensive and capital-intensive groups for our difference-in-differences (DiD) analysis. According to the 2016 BLS Foreign-born Workers report, foreign-born workers are more likely to work in labor-intensive occupations, while native-born workers are more likely to be employed in high-skilled occupations.[2] Our DiD regression framework thus compares the changes in labor-shortage exposure of labor-intensive firms with the changes in labor-shortage exposure of capital-intensive firms three years before and after

[1] See https://www.bls.gov/news.release/archives/forbrn05182017.pdf.

[2] See https://www.bls.gov/news.release/archives/forbrn_05182017.pdf. For example, a higher portion of foreign-born workers are employed in *service, production*, *transportation and material moving*, *natural resources*, *construction*, and *maintenance* occupations than native-born workers, while native-born workers are more likely to work in *management*, *professional and related*, and *sales and office* occupations.

2017 (i.e., from 2014 to 2019; we avoid the COVID-19 crisis for this analysis). Because labor-intensive firms are more dependent on the foreign labor supply than capital-intensive firms, they should be more affected by the 2017 immigration policy reforms. The DiD regression specification is as follows:

$$Y_{i,t} = \beta_1 IMMI\ Post_t \times Labor\ Intensive_{s,i} + \beta_2 Controls_{i,t-1} + \omega_i + \mu_t + \epsilon_{i,t} \tag{1}$$

We further use the following dynamic DiD regression framework to identify the exact timing of the effect:

$$Y_{i,t} = \sum_{j=-2}^{2} \beta_j Labor\ Intensive_{s,i} \times Year_j + \beta_{j+1} Controls_{i,t-1} + \omega_i + \mu_t + \epsilon_{i,t} \tag{2}$$

In Equation 1, Y denotes the labor-shortage exposure of firm i in year t, *IMMI Post* is an indicator that equals one if year t is 2017 or after, and equals zero otherwise. Following prior literature (see, e.g., **?**; **?**; **?**), we use the labor-capital ratio to capture the relative amount of labor and capital used in a firm's production process. A high (low) labor-capital ratio indicates that the firm relies more heavily on labor inputs (capital inputs) in its production process. The labor-capital ratio is calculated as the number of employees divided by the value of fixed assets. We classify firm i as labor intensive if the firm's labor-capital ratio is above the median value of the firms in the same 2-digit SIC industry s in year 2016. The indicator, *Labor Intensive*, equals one for such labor intensive firms and equals zero otherwise.[3] The coefficient of interest, β_1, captures the differential effect of the immigration policy reforms on firm-level labor-shortage exposure between labor-intensive firms and capital-intensive firms in the same industry. We further control for lagged firm size as well as firm fixed effects ω_i and year fixed effects μ_t. We expect β_1 to be significantly positive if our measure reflects firm-level labor-shortage exposure. Equation 2 is similar to Equation 1 except that we replace the post indicator (*IMMI Post*) with a series of year indicators to allow for the differential effect to vary across the sample years, with year 2014 being the reference year in the dynamic DiD regressions. The results are reported in Table B1.

[Please insert Table B1 about here]

Columns 1, 3, and 5 control for firm and year fixed effects, while columns 2, 4, and 6 further replace year fixed effects with industry-by-year fixed effects to account for time-varying industry characteristics. All specifications control for one-year-lagged firm size. Columns 1 and 2 show that the coefficient estimates on the DiD term, *IMMI Post*×*Labor Intensive*, is positive and statistically significant at the 5% level. The economic magnitude is sizeable. For example, column 1 shows that compared with the control firms, the treated firms experience an average increase in labor-shortage exposure of 0.012 post-IMMI, which is equivalent to one fifth of the sample mean of 0.062. Thus, consistent with our expectation, the results suggest that after the enactment of the immigration policy reforms, relative to the control firms (capital-intensive firms), the treated firms (labor-intensive firms) experience a significant increase in labor-shortage exposure.

[3] Since we develop a firm-level labor-shortage measure, partitioning firms into labor-intensive and capital-intensive groups by each industry helps mitigate the concern that some industry-level confounding factors may drive the differential effect of the immigration policy reforms on labor-intensive firms relative to capital-intensive firms.

Next, we explore the heterogeneity in the positive effect of tightened immigration policies on corporate labor-shortage exposure. Specifically, for each industry, instead of partitioning firms into high- and low-labor-capital-ratio groups, we divide the sample firms into quartiles based on a firm's labor-capital ratio in 2016. We then use a regression specification similar to Equation 1 to test which quartile exhibits largest effect. We expect that firms with the highest labor intensity quartile should be most exposed to labor-shortage issues after the immigration policy reforms. Columns 3-4 report the results. We find that relative to capital-intensive firms in the lowest labor-capital-ratio quartile, the tightened immigration policies have insignificant effect on labor-shortage exposure of firms in quartiles 2-3, while firms in quartile 4 experience a sizable and statistically significant (at the 1% level) increase in corporate labor-shortage exposure. These results are consistent with our expectation that the positive effect of the immigration policy reforms on labor-shortage exposure mainly concentrates in firms with high labor intensity within an industry.

Finally, we investigate whether potential nonparallel trends exist before the reforms. Columns 5-6 report the results of estimating Equation 2. We find that the coefficient estimates of the interaction terms are all insignificantly different from zero for the years before the event year ($Year_0$) across the regression specifications in both columns. The positive effect on corporate labor-shortage exposure can only be observed after the event year, with the largest effect occurring in the year immediately after the event year. These findings suggest that the increase in firm-level labor-shortage exposure for the labor-intensive firms relative to the control firms is unlikely to be driven by pre-event nonparallel trends in labor-shortage exposure but is most likely caused by the tightened immigration policies that reduce foreign labor supply.

To summarize, the results based on the 2017 immigration policy reforms further confirm the validity of the measure of firm-level labor-shortage exposure. The U.S. immigration reforms in 2017 shrink foreign labor supply, which directly affects labor-intensive firms because such firms rely heavily on foreign-born workers. Thus, we observe a significant increase in labor-shortage exposure for such firms after the reforms.

B1.2 Federal Pandemic Unemployment Compensation (FPUC) Program

In this section, we use the Federal Pandemic Unemployment Compensation (FPUC) program introduced under the CARES Act during COVID-19 to further validate our measure of firm-level labor-shortage exposure. From April to July 2020, FPUC provided a $600/week supplement to standard unemployment benefits. As documented by **?**, roughly 76% of UI-eligible workers received benefits exceeding their prior earnings, with a median replacement rate of 145%. The implementation of the FPUC program therefore significantly reduced the incentive for low-wage workers to remain employed, generating a substantial and plausibly exogenous labor supply shock. With reduced labor supply, we expect the FPUC program to increase labor shortages for firms, particularly those employing a higher proportion of low-wage workers.

To formally investigate the effect of the FPUC program on firms' labor-shortage exposure, we implement a DiD regression framework as follows:

$$LS\ Exposure_{i,q+1} = \beta_1 Low\ Wage_i \times Post_q + \beta_2 Controls_{i,q-1} + \omega_i + \mu_q + \epsilon_{i,q} \tag{3}$$

In Equation 3, the dependent variable, *LS Exposure*, represents the labor-shortage exposure

of firm i in year-quarter *q+1*. We then partition our sample firms into low-wage and high-wage groups for our DiD analysis. The independent variable, *Low Wage*, is an indicator variable that equals one if a firm operates in a 6-digit NAICS industry that falls in the bottom quartile of average weekly wages in 2019, and zero otherwise. According to the labor wage statistics from Quarterly Census of Employment and Wages (QCEW) of U.S. Bureau of Labor Statistics (BLS), workers at firms in low-wage group earned an average of $947 per week in 2019, compared to $2,314 for others. Thus, these "low-wage" firms are arguably more affected by the the FUPC program and serve as our treatment firms. Using a balanced quarterly panel from 2019Q4 to 2020Q3, we define the *Post* period as 2020Q2–Q3. *Post* is an indicator that equals one if the year-quarter q is 2020Q2 or 2020Q3, and equals zero if the year-quarter is 2019Q4 or 2020Q1.

The coefficient of interest, β_1, captures the differential effect of the FPUC program on firm-level labor-shortage exposure between low-wage and high-wage firms. We further control for lagged firm size as well as firm fixed effects ω_i and year-quarter fixed effects μ_q. We expect β_1 to be significantly positive if our measure reflects firm-level labor-shortage exposure. The results are reported in Panel A of Table B2.

[Please insert Table B2 about here]

In Columns 1–2, we use *LS Exposure* as the dependent variable. In Columns 3–4, we instead use an indicator variable, *I(LS)*, which equals one if the firm's labor-shortage exposure is greater than zero and zero otherwise. Columns 1–2 show that the coefficient estimates on the DiD term, *Low Wage*×*Post*, are positive and statistically significant at the 1% level. We find similar results in Columns 3–4, indicating that low-wage firms are significantly more likely to experience labor shortages following the implementation of the FPUC program. The estimated effects are economically meaningful. For example, Columns 2 and 4 suggest that, relative to control firms, low-wage firms exhibit an increase in labor-shortage exposure of 0.042 (compared to a sample mean of 0.062) and a 3.9% higher likelihood of being exposed to labor shortages, respectively, after the FPUC program was introduced.

In Panel B of Table B2, we further explore the dynamic effects of the FPUC program on low-wage firms' labor-shortage exposure. Specifically, we replace the *Post* indicator in Equation 3 with a series of year-quarter indicators: Pre_1, $Start_0$, and $Post_1$, which equal one if the year-quarter is 2020Q1, 2020Q2, and 2020Q3, respectively, and zero otherwise. The results in Panel B show that the coefficient estimate on the interaction term *Low Wage*×Pre_1 is statistically insignificant. A sharp and persistent increase in *LS Exposure* is observed only during and after the event quarters, supporting the parallel-trends assumption of the DiD regressions. Similar patterns are observed for *I(LS)*.

In summary, the results in this section support our expectation that the enactment of the FPUC program significantly increased low-wage firms' labor-shortage exposure. These validation tests provide further evidence of the reliability of the firm-level labor-shortage exposure measure.

Table B1. Validation: The Effect of the 2017 Tightened U.S. Immigration Policy on Firms' Labor-Shortage Exposure

This table presents a validation test of our firm-level labor-shortage exposure measure. Columns 1-4 report the difference-in-differences (DiD) regression results using the 2017 tightened U.S. immigration policy as an exogenous shock on local firms' labor-shortage exposure. *IMMI Post* is an indicator variable that equals one if the year is 2017 or after and equals zero otherwise. *Labor Intensive* is an indicator variable that equals one if a firm's labor capital ratio (emp/ppent) based on the 2016 value is higher than (2-digit SIC) industry median and equals zero otherwise. *Labor Intensive Q2(Q3/Q4)* is an indicator variable that equals one if a firm's labor-capital ratio (emp/ppent) based on the 2016 value is in the second-quartile (third-quartile/fourth-quartile) in its (2-digit SIC) industry and equals zero otherwise. Columns 5-6 report the dynamic DiD regression results that investigate the timing of the effect of the tightened immigration policy on firms' labor-shortage exposure. $Year_j$ is an indicator variable that equals one if the year is the jth year relative to the event year (year zero, which is 2017) and equals zero otherwise. All specifications include firm fixed effects and year (industry-year) fixed effects. We also control for firm size in all specifications. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES			LS Exposure			
IMMI Post×Labor Intensive	**0.012****	**0.012****				
	(0.005)	**(0.005)**				
IMMI Post×Labor Intensive^{Q2}			0.001	0.001		
			(0.006)	(0.006)		
IMMI Post×Labor Intensive^{Q3}			0.002	0.002		
			(0.007)	(0.007)		
IMMI Post×Labor Intensive^{Q4}			**0.026*****	**0.026*****		
			(0.008)	**(0.008)**		
Labor Intensive×Year$_{-2}$					-0.003	-0.003
					(0.006)	(0.005)
Labor Intensive×Year$_{-1}$					0.011	0.006
					(0.006)	(0.006)
Labor Intensive×Year$_0$					**0.011**	**0.008**
					(0.007)	**(0.007)**
Labor Intensive×Year$_{+1}$					**0.023*****	**0.024*****
					(0.009)	**(0.008)**
Labor Intensive×Year$_{+2}$					**0.012**	**0.009**
					(0.009)	**(0.009)**
Firm Size $_{t-1}$	0.002	0.005**	0.003	0.005**	0.002	0.005**
	(0.003)	(0.002)	(0.003)	(0.002)	(0.003)	(0.002)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No
Industry-Year FE	No	Yes	No	Yes	No	Yes
Obs.	12,980	12,980	12,980	12,980	12,980	12,970
Adj. R2	0.468	0.502	0.469	0.503	0.468	0.503

Table B2. Validation: The Effect of Federal Pandemic Unemployment Compensation (FPUC) Program on Firms' Labor-Shortage Exposure

This table presents a validation test of our firm-level labor-shortage exposure measure. In Panel A, we report the difference-in-differences (DiD) regression results using the Federal Pandemic Unemployment Compensation (FPUC) Program as an exogenous shock on low-wage firms' labor-shortage exposure. The dependent variable *LS Exposure* is a firm's labor-shortage exposure in a year-quarter; *I(LS)* is an indicator that equals one if the firm's labor-shortage exposure is greater than zero, otherwsie equals zero. The independent variable *Post* is an indicator variable that equals one if the year-quarter is 2020Q2 or 2020Q3, and equals zero if the year-quarter is 2019Q4 or 2020Q1. *Low Wage* is an indicator variable that equals one if a firm operates in a 6-digit NAICS industry that falls within the bottom quartile of average weekly wages in 2019, otherwise equals zero. In Panel B, we report the dynamic DiD regression results that investigate the timing of the effect of the FPUC program on firms' labor-shortage exposure. The independent variable Pre_1 ($Post_1$) is an indicator variable that equals one if the year-quarter is one quarter before (after) the event year-quarter (2020Q2), otherwise equals zero. $Start_0$ is an indicator variable that equals one if the year-quarter is the event year-quarter (2020Q2), otherwise equals zero. All specifications include firm and year-quarter fixed effects. Columns 2 and 4 further control for firm size. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. All financial variables are winsorized at the 1st and 99th percentiles. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. The effect of the FPUC program on firms' labor-shortage exposure

	(1)	(2)	(3)	(4)
VARIABLES	LS Exposure $_{q+1}$		I (LS) $_{q+1}$	
Low Wage×Post	**0.042*** **	**0.042*** **	**0.038* **	**0.039** **
	(0.011)	**(0.011)**	**(0.020)**	**(0.020)**
Firm Size $_{q-1}$		0.004		0.009
		(0.005)		(0.011)
Firm FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Obs.	9,640	9,640	9,640	9,640
Adj R2	0.310	0.310	0.242	0.242

Panel B. Dynamic difference-in-differences tests

	(1)	(2)	(3)	(4)
VARIABLES	LS Exposure $_{q+1}$		I (LS) $_{q+1}$	
Low Wage×Pre_1	-0.004	-0.004	-0.002	-0.002
	(0.013)	(0.013)	(0.026)	(0.026)
Low Wage×$Start_0$	**0.041*** **	**0.041*** **	**0.045* **	**0.046* **
	(0.014)	**(0.014)**	**(0.026)**	**(0.026)**
Low Wage×$Post_1$	**0.039*** **	**0.039*** **	**0.029**	**0.030**
	(0.014)	**(0.014)**	**(0.026)**	**(0.026)**
Firm Size	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Obs.	9,640	9,640	9,640	9,640
Adj R2	0.310	0.310	0.242	0.242

Appendix C Addressing Endogeneity

In this section, we construct two Bartik shift-share instruments (**?**; **?**) to further address the endogeneity concern that unobserved time-varying firm-level factors may correlate with both firm-level labor-shortage exposure and future firm performance. The Bartik instrument aims to identify the treatment effect by measuring the differential impact of common shocks on units with distinct predetermined exposures. For instance, **?** instruments a county's employment rate by interacting the nationwide industry employment rate with the county's preexisting industry shares. Inspired by this, we utilize the granular information on a firm's segment sales from the Compustat Segments database to construct an instrument for the firm-level labor-shortage exposure.

Specifically, we construct the instrument $Bartik\ IV1_{\mathrm{i,t}}$ by summing the products of $LS\ Exposure_{\mathrm{j,t}}{}^{\mathrm{Ind\text{-}Avg}}$, the average labor-shortage exposure for industry j in year t,[4] and $Segment\ Share_{\mathrm{j,i,t\text{-}1}}$, the predetermined segment sales share of firm i in industry j in year *t-1*. We then instrument $LS\ Exposure_{\mathrm{i,t}}$ with $Bartik\ IV1_{\mathrm{i,t}}$ in two-stage least squares (2SLS) regressions. That is, we use aggregate industry-year labor shortage conditions, weighted by a firm's predetermined segment sales shares, to predict its labor-shortage exposure. The rationale is that preexisting segment sales shares reflect the firm's ex-ante exposure to specific industries. When labor shortages affect these industries, firms with higher preexisting sales in those industries are expected to be more significantly impacted. This approach helps isolate variation in labor-shortage exposure from firm-specific characteristics. Panel A of Table C1 reports the results.

[Please insert Table C1 about here]

Column 1 presents the first-stage regression results, where $LS\ Exposure_{\mathrm{i,t}}$ is regressed on $Bartik\ IV1_{\mathrm{i,t}}$, with controls for firm size and firm and year fixed effects. As anticipated, we find that the instrument is positively and significantly related to firm labor-shortage exposure, with Cragg-Donald Wald F statistic exceeding 288. Columns 2 to 8 present the second-stage results using the instrumented labor-shortage exposure, $LS\ \widehat{Exposure}_{\mathrm{i,t}}$. Specifically, Columns 2-3 explore the effects of instrumented labor-shortage exposure on future firm performance, while Columns 4-8 examine corporate policy responses to labor-shortage issues. Consistent with our prior findings, the results suggest that firm-level labor-shortage exposure indeed negatively affects future firm performance, and firms seek to address labor-shortage issues by replacing labor with capital and process innovations.

The second-stage estimates are much larger than the OLS estimates likely because our Bartik instrument interacts common industry shifts in labor shortages with a firm's predetermined industry segment sales shares. The resulting 2SLS estimates effectively identify a local average treatment effect (LATE) for firms whose ex-ante exposure is most moved by these shocks. Economically, such firms plausibly face more binding hiring frictions, so their adjustment responses can be larger than the unconditional average effect estimated by OLS. Moreover, measurement error in the *LS Exposure* measure (e.g., due to disclosure-related noise) can attenuate OLS estimates toward zero, while instrument corrects that attenuation.

To make the comparison between OLS and 2SLS estimates transparent, in addition to reporting first-stage strength diagnostics, we report the economic magnitudes of the estimated effects using an interquartile change (or one-standard-deviation increase) in *LS Exposure* for OLS

[4] We use only the labor-shortage exposures of single-segment firms in industry j to calculate the industry's average exposure.

regressions and an interquartile change (or one-standard-deviation increase) in $LS\ \widehat{Exposure}$ (i.e., the instrumented measure) for the 2SLS regressions. As shown in Table C2 in Appendix C, across our core outcome variables, the signs align and the magnitudes scale sensibly: the economic magnitudes of the 2SLS effects are larger but of the same order of magnitude as OLS when expressed in interquartile or one-standard-deviation terms.

We construct one further instrument in the spirit of **?**; we interact $LS\ Exposure_{j,t}{}^{M}$, the average labor-shortage exposure for *labor-shortage-mentioning firms* in industry j (to which firm i belongs) and year t, with $LS\ Exposure_{j,t\text{-}1}{}^{Fraction}$, the predetermined fraction of firms exposed to labor shortages in industry j and year *t-1*. The idea behind this instrument is that a firm is more likely to be affected by industry-wide labor shortages if the intensive margin of labor-shortage exposure in its industry is high currently and labor-shortage issues were pervasive in the industry previously. The first-stage results in Column 1 of Panel B in Table C1 confirm this hypothesis. Importantly, the second-stage results, reported in Columns 2-8, indicate that the negative effects of firm-level labor-shortage exposure on future stock performance remain robust. Moreover, we continue to observe the substitution effects between labor and capital and process innovations.

In summary, the results from both sets of 2SLS regressions yield qualitatively similar findings: labor shortages decrease firms' future performance and motivate them to replace labor with capital and process innovations.

Table C1. Addressing Endogeneity Concerns: Bartik Shift-Share Instrumental-Variable Analyses

The table reports the results from Bartik shift-share instrumental-variable analyses. In Panel A, we construct an instrument *Bartik IV1*$_{i,t}$ by summing the products of *LS Exposure*$_{j,t}^{\text{Ind-Avg}}$ and *Segment Share*$_{s,i,t-1}$ for each firm *i* in industry *j* and year *t*. *LS Exposure*$_{j,t}^{\text{Ind-Avg}}$ is the average labor-shortage exposure for industry *j* (to which the segment *s* of firm *i* belongs) in year *t*. *Segment Share*$_{s,i,t-1}$ is the predetermined sales share of segment *s* in firm *i* in year *t-1*. In Panel B, *Bartik IV2* is the product of *LS Exposure*$_{j,t}^{M}$, the average labor-shortage exposure for *LS-mentioning firms* in industry *j* and year *t*, and *LS Exposure*$_{j,t-1}^{Fraction}$, the fraction of firms exposed to labor shortages in industry *j* and year *t-1*. In all specifications, we include control variables, firm fixed effects, and year fixed effects. Table A1 in Appendix A provides detailed variable definitions. Robust standard errors clustered at the firm level are provided in parentheses. ***, **, and * correspond to statistical significance at the 1, 5, and 10 percent levels, respectively.

Panel A. Industry-year labor shortage exposure weighted by preexisting firm segment sales shares

VARIABLES	(1) LS Exposure $_t$	(2) Stock Return$_{t+1}$	(3) Ind-adj.ROA$_{t+1}$	(4) CAPEX$_{t+1}$	(5) R&D$_{t+1}$	(6) $\triangle$Employees/AT$_{t+1}$	(7) Process Claim Share$_{t+1,t+3}$	(8) CW Process Patents/AT$_{t+1,t+3}$
	First Stage	*Second Stage*						
Bartik IV1	**0.303*** **							
	(0.034)							
LS $\widehat{\text{Exposure}}$		**-1.011*** **	**-0.242*** **	**0.017**	**0.325*** **	**-0.004*** **	**0.999*** **	**0.192** **
		(0.275)	**(0.064)**	**(0.025)**	**(0.064)**	**(0.001)**	**(0.263)**	**(0.070)**
Cragg-Donald Wald F Stat	288.68							
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	35,566	35,566	35,495	31,457	19,731	31,554	16,343	16,343
Adj. R2	0.380	-0.052	-0.059	-0.000	-0.251	-0.165	-0.648	-0.076

Panel B. Industry-year labor-shortage exposure (LS-mentioning firms) multiplied by preexisting fraction of LS-mentioning firms in industry

VARIABLES	(1) LS Exposure $_t$	(2) Stock Return$_{t+1}$	(3) Ind-adj.ROA$_{t+1}$	(4) CAPEX$_{t+1}$	(5) R&D$_{t+1}$	(6) $\triangle$Employees/AT$_{t+1}$	(7) Process Claim Share$_{t+1,t+3}$	(8) CW Process Patents/AT$_{t+1,t+3}$
	First Stage	*Second Stage*						
Bartik IV2	**0.468*** **							
	(0.036)							
LS $\widehat{\text{Exposure}}$		**-0.597*** **	**-0.153*** **	**0.046*** **	**0.244*** **	**-0.002*** **	**0.952*** **	**0.206*** **
		(0.140)	**(0.035)**	**(0.016)**	**(0.048)**	**(0.001)**	**(0.259)**	**(0.073)**
Cragg-Donald Wald F Stat	930.41							
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	30,757	30,757	30,712	26,995	17,016	27,072	14,460	14,460
Adj. R2	0.411	-0.006	-0.009	-0.031	-0.131	-0.038	-0.586	-0.097

Table C2. Comparison of Coefficient Estimates and Economic Magnitudes between OLS and 2SLS Regressions

The table compares the coefficient estimates and economic magnitudes of the estimated effects between the baseline OLS regressions and the 2SLS regressions. In Panel A, we report the coefficient estimates on *LS Exposure* from OLS regressions and *LS* $\widehat{Exposure}$ from 2SLS regressions (Panel A of Table C1). In Panel B (C), we compare the economic magnitudes of the estimated effects using an interquartile (one-standard-deviation) increase in *LS Exposure* for OLS and an interquartile (one-standard-deviation) increase in *LS* $\widehat{Exposure}$ for 2SLS (Panel A of Table C1). Table A1 in Appendix A provides detailed variable definitions.

VARIABLES	(1) Stock Return$_{t+1}$	(2) Ind-adj.ROA$_{t+1}$	(3) CAPEX$_{t+1}$	(4) R&D$_{t+1}$	(5) $\triangle$Employees/AT$_{t+1}$	(6) Process Claim Share$_{t+1,t+3}$	(7) #CW Process Patents/AT$_{t+1,t+3}$
Panel A. Regression coefficients from OLS/IV							
LS Exposure	-0.092	-0.017	0.008	0.018	-0.0002	0.058	0.008
LS $\widehat{Exposure}$	-1.011	-0.242	0.017	0.331	-0.004	0.999	0.192
Panel B. Economic magnitudes (in percentage point) of an interquartile increase in LS Exposure (0.060)/LS $\widehat{Exposure}$ *(0.015)*							
OLS	-0.55	-0.10	0.05	0.11	-0.001	0.35	0.05
IV	-1.52	-0.36	0.03	0.50	-0.006	1.50	0.29
Panel C. Economic magnitudes (in percentage point) of one-SD increase in LS Exposure (0.149)/LS $\widehat{Exposure}$ *(0.018)*							
OLS	-1.37	-0.25	0.12	0.27	-0.003	0.86	0.12
IV	-1.82	-0.44	0.03	0.60	-0.007	1.80	0.35